2017 13th IEEE Conference on Automation Science and Engineering (CASE)
Xi'an, China, August 20-23, 2017

Study of effects of high-altitude environments on Multicopter and Fixed-Wing UAVs' Energy Consumption and Flight Time

Juan Augusto Paredes, Carlos Saito, Monica Abarca, Francisco Cuellar
Unmanned Aerial Systems Laboratory and Technological Innovation Group - Pontifical Catholic University of Peru (PUCP), Lima, Peru
japaredes@pucp.pe, csaito@pucp.pe, monica.abarca@pucp.pe, cuellar.ff@pucp.pe

***Abstract* —The effects of low air density (characteristic at high-altitude environments) on small-sized Unmanned Aerial Vehicles (UAVs) are studied in this paper. Flight data from a fixed-wing aircraft and a quadcopter (in representation of multicopters) is examined to substantiate the influence high altitude conditions have on small UAVs' flight. By using aeronautical theoretical background to construct a hypothesis concerning how altitude affects the energy consumption and, consequently, flight time of both types of UAV, a qualitative analysis of flight missions held at different altitudes will be performed to find a correlation between the energy consumption and target altitude to determine if low altitude flight missions could be used to predict consumption at high-altitude environments. Finally, guidelines abstracted from these flight experiences will be given to help in the future design of UAVs with high altitude applications.**

***Keywords*—Aerospace Technology, Unmanned Aerial Vehicles, Airworthiness, Aircraft Design, High-Altitude Missions, Energy Consumption, Flight Time, Autonomous Vehicle Navigation**

I. MOTIVATION

The effects of altitude on performance of regular-sized certified aircraft models in service are thoroughly studied to determine optimal operation altitude and properties such as service and absolute ceiling. This manner of thoroughness, however, is not required from small unmanned aircraft, as declared in the Small Unmanned Aircraft Regulations (Part 107), which states that "the Federal Aviation Administration (FAA) does not require small Unmanned Aerial Systems (UASs) to comply with current agency airworthiness standards or obtain aircraft certification" [1]. The European Aviation Safety Agency (EASA) proposed a similar approach by establishing standard safety rules for all civil drone operations and offering a classification system for commercially available ones that provides required pilot qualifications for their handling and permitted mission parameters for each class [2]; this set of rules allow manufacturers and researchers to have a lot of design freedom. Nevertheless, this limits the required information that companies need to provide about operation in multiple environmental situations.

It is favorable for Unmanned Aerial Vehicle (UAV) manufacturing companies to offer the technical specifications of their Ready-To-Fly airframes since these aren't meant to be customized by users and this information allows clients to be aware of the limitations of the models they buy. Among the drones that feature flight time and service ceiling (or some measure of altitude at which the model can fly) in their descriptions are the Matrice 600 Pro [3] from DJI and the MD4-3000 [4] from Microdrones (coincidentally, these are said to have high altitude operating capabilities). A well-known fixed-wing UAV in the market, the eBee [5] from senseFly, has only listed its flight time among its specifications, although its high-altitude capabilities can be inferred from the Matterhorn mountain mapping project, in which a team of engineers flew eBees at a maximum altitude of 4707 Meters Above Sea Level (MASL) [6].

Companies and institutes have started using UASs in high-altitude environments (above 3500 MASL), mainly for imaging and monitoring purposes. For example, in Switzerland, as it was briefly mentioned above, a senseFly engineering team used the eBee flying platform to take images and create a 3D digital model of the Matterhorn mountain in the Swiss Alps [6]. Other high-altitude 3D reconstruction ventures were held in Peru [7]. Volcanic monitoring has also been identified as a potential application for which small UASs are especially suited. Studies and flight tests were held at Italy [8], Japan [9] and Peru [10], thus acknowledging the need of flying equipment capable of thriving in environments with low air density. Therefore, an assessment of the effects this has on energy consumption and, subsequently, flight time in comparison to test site performance would be helpful to establish guidelines for small unmanned aircraft design under these circumstances.

The objective of this paper is to analyze how high-altitude conditions affect lift generation, energy consumption and, therefore, flight time, and, as a result, to determine if energy consumption of UAVs at different altitudes can be predicted from flight test data from lower altitudes. Another aim of this document is to provide guidelines for future UAV designs obtained from several flight experiences. Section I introduces the motivation behind the present study and relevant research background. Section II exposes the theoretical background that will help formulate a hypothesis about altitude effect on UAVs. Section III presents the frames and models used for flight tests and the testing sites and flight conditions corresponding to the missions held. Section IV displays the information attained from the flight tests and presents a discussion of these results and their link to altitude changes as

well as guidelines for designers to consider when building a UAV for high-altitude operations. Finally, section V features the conclusions obtained from this study and future work. The symbols that will be used in this document and their definition are displayed in Table I.

TABLE I
LIST OF SYMBOLS

Symbol	Definition
L	Lift [N]
D	Drag [N]
C_L	Lift Coefficient
C_D	Drag Coefficient
ρ	Air density [kg/m^3]
V	Air velocity [m/s]
S	Surface Area [m^2]
Th	Thrust [N]
Q	Torque [N.m]
C_{Th}	Thrust Coefficient
C_Q	Torque Coefficient
n	Rotation frequency [rpm]
P_R	Power required (W)
Di	Propeller Diameter (m)
P	Power (W)
H	Height above sea level [m]
T	Temperature [K]
λ	Troposphere temperature gradient ($\frac{dT}{dH} = -0.0065\ \frac{K}{m}$)
g	Acceleration of gravity at sea level (9.81 m/s^2)
R	Specific gas constant for air (287.05 m^2 /s^2 K)
W	Weight [N]
γ	Angle between velocity vector and horizontal plane [rad]
A_R	Rotor area [m^2]
N_R	Multicopter number of rotors

II. THEORETICAL BACKGROUND

An airfoil requires air to flow through it at a given velocity to generate lift. Under these circumstances, a force opposing the airflow direction called drag is also present, which must be overcome for the airfoil to accelerate. Both wings and propellers are composed by airfoils and the Lift and Drag they experience can be represented through these equations:

$$L = \frac{1}{2}\,\rho\,V^2\,S\,C_L, \quad (1)$$

$$D = \frac{1}{2}\,\rho\,V^2\,S\,C_D, \quad (2)$$

where C_L and C_D are non-dimensional coefficients that depend on aerodynamic properties that characterizes a chosen design of wings and propellers. It should be noted that C_L can be increased by changing an airfoil's angle of attack.

Airplanes and rotorcraft depend on similar aerodynamic principles for lift generation. However, the mechanics through which both keep afloat differ greatly. Airplanes require a stream of wind at a certain velocity to pass through their wings to generate enough lift to oppose their weight, which explains why these are kept in constant forward movement through propulsion systems, like turbines or propellers. In contrast, multirotors can hover in place since the relative airspeed their propellers need to generate lift is provided by the rotation of their motors; this requires the propellers to keep rotating to maintain a certain altitude, whereas small fixed-wing UAVs can keep flying without propeller rotation as long as relative airspeed is kept constant. Both types of UAV are shown in Figure 1.

Fig. 1.- UAV frames. [Left] Fixed-wing frame. [Right] Rotorcraft frame.

Since small scaled fixed-wing aircraft and multirotors both use propellers, concepts about propeller performance will be reviewed using [11] as reference. Propeller thrust can be expressed through the following equation:

$$Th = C_{Th}\,\rho\,n^2\,Di^4. \quad (3)$$

The aerodynamic drag generated by rotating the propeller applies a torque on the rotor, which must be overcome for it to keep spinning. This torque can be represented through this expression:

$$Q = C_Q\,\rho\,n^2\,Di^5, \quad (4)$$

which, means the required power can be evaluated through this representation:

$$P_R = 2\,\pi\,n\,Q = 2\,\pi\,C_Q\,\rho\,n^3\,Di^5. \quad (5)$$

C_{Th} and C_Q serve the same purpose as C_L and C_D since both encompass several aerodynamic properties of the propeller blades.

Air density has been prominent in each of the equations that has been presented, and rightfully so since air circulation dynamics depend substantially on the cohesion between air particles. This property diminishes as altitude increases due to pressure loss. Assuming standard atmosphere conditions and operations within the troposphere (linear temperature decrease as altitude increases), according to [12], change in air density due to height can be represented as follows:

$$\frac{\rho}{\rho_0} = \left(1 + \frac{\lambda\,H}{T_0}\right)^{-\left(1+\frac{g}{R\,\lambda}\right)} \cong \left(1 - 2.2558\,.\,10^{-5}\,H\right)^{4.2577}, \quad (6)$$

in which the 0 subindices represent variables at sea level conditions, T_0 = 288.15 K (15° C) and ρ_0 = 1.225 kg/m^3. Figure 2 shows how relative air density changes with altitude increase. It is interesting to notice that at 7000 MASL air density has decreased to a value below 50% of the one at sea level.

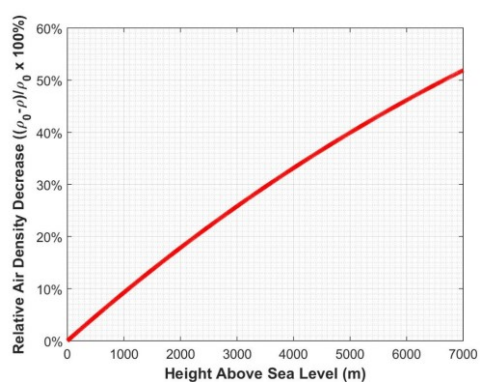

Fig. 2.- Decrease in air density relative to sea level with altitude increase.

Considering equations (1) and (3), loss in air density reduces a UAV's flying capability. This can be overcome by changing the frame design and increasing other factors, like wing area (S) or propeller diameter (D_i) or by changing aerodynamic characteristics. However, if it isn't viable to

implement these modifications, the only available option to increase lift and maintain current altitude, is to increase the aircraft forward velocity in fixed-wing platforms (by increasing its motor's RPM) and to augment the rotational velocity of all the rotors in the case of multicopters. As equation (5) states, this would cause a raise in the required power in a cubic fashion, thus overshadowing the decrease caused by the reduction in air density, resulting in elevated energy consumption.

Most fixed-wing UAVs rely on propellers for propulsion and, indirectly, lift generation. According to [13], under symmetric flight conditions and using a coordinates system aligned with an oncoming airstream (as shown in Figure 3), assuming the rotor is almost perfectly aligned with the aircraft, the equations of motion that govern its movement are represented by the following equations:

$$\mathrm{Th} - \mathrm{D} - \mathrm{W}\ \sin\gamma = \frac{\mathrm{W}}{\mathrm{g}}\frac{\partial \mathrm{V}}{\partial \mathrm{t}}, \tag{7}$$

$$\mathrm{L} - \mathrm{W}\ \cos\gamma = \frac{\mathrm{W}}{\mathrm{g}}\ \mathrm{V}\ \frac{\partial \gamma}{\partial \mathrm{t}}. \tag{8}$$

As equation (7) shows, the thrust exerted by the rotor will oppose drag and a component of weight, and be responsible for velocity changes. This variable, along with the angle of attack, increase lift, which opposes the larger weight component and balances change in angle γ, as shown in (8). Following suit, as stated in [14], in steady symmetric flight, in which V and angle γ are kept constant, equation (7) and (8) change in the following manner:

$$\mathrm{Th} - \mathrm{D} - \mathrm{W}\ \sin\gamma = 0, \tag{9}$$

$$\mathrm{L} - \mathrm{W}\ \cos\gamma = 0. \tag{10}$$

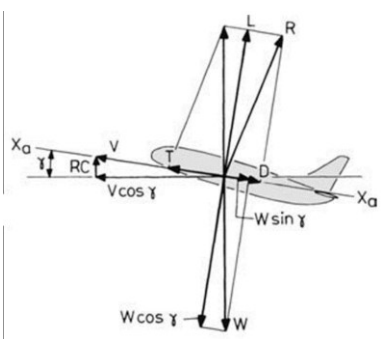

Fig. 3.- Airplane in steady symmetric flight. Obtained from [14].

By rearranging equations (1), (2) and (10), we can establish definitions for velocity, drag and power required as follows:

$$\mathrm{V} = \sqrt{\frac{\mathrm{W}\ \cos\gamma}{\frac{1}{2}\rho\ \mathrm{S}\ \mathrm{C_L}}} \cong \sqrt{\frac{\mathrm{W}}{\frac{1}{2}\rho\ \mathrm{S}\ \mathrm{C_L}}}, \tag{11}$$

$$\mathrm{D} = \frac{\mathrm{C_D}}{\mathrm{C_L}}\ \mathrm{W}\ \cos\gamma \cong \frac{\mathrm{C_D}}{\mathrm{C_L}}\ \mathrm{W}, \tag{12}$$

$$\mathrm{P_R} = \mathrm{DV} = \frac{(\mathrm{W}\ \cos\gamma)^{3/2}}{\sqrt{\frac{1}{2}\rho\ \mathrm{S}\frac{\mathrm{C_L}^3}{\mathrm{C_D}^2}}} \cong \frac{\mathrm{W}^{3/2}}{\sqrt{\frac{1}{2}\rho\ \mathrm{S}\frac{\mathrm{C_L}^3}{\mathrm{C_D}^2}}}. \tag{13}$$

Both the required velocity and power grow asymptotically to $\rho^{-1/2}$, which means that both increase with altitude raise. However, the drag equation in (12) is not affected whatsoever by ρ, implying that, while maintaining steady symmetric flight, the effect of drag on the aircraft remains the same at different altitudes. Moreover, from (13), it can be inferred that increasing S and attaining an optimal value of $\mathrm{C_L}^3/\mathrm{C_D}^2$ by tuning the aerodynamic design of the aircraft would lead to a more energy efficient flight system.

Multicopters' dynamic systems behave differently under the effects of rotor thrust and weight, as shown in Figure 4. Each rotor exerts thrust (Th) and torque (M), perpendicular to the plane in which each propeller rotates. In hover conditions, the main equation of motion that governs multicopters' vertical motion is the following:

$$\sum_{\mathrm{i}=1}^{\mathrm{N_R}} \mathrm{Th_i} - \mathrm{W} = \mathrm{N_R}\ \mathrm{Th} - \mathrm{W} = \frac{\mathrm{W}}{\mathrm{g}}\frac{\partial \mathrm{H}}{\partial \mathrm{t}} = 0. \tag{14}$$

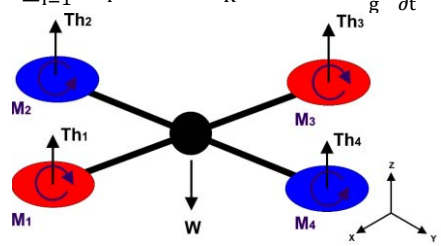

Fig. 4.- Free body diagram of quadcopter.

Applying momentum theory for rotatory wing vehicles as is specified in [15] yields an equation to represent the power required by each rotor to keep a multicopter hovering at a specific altitude, which is:

$$\mathrm{P_R} = \frac{(\mathrm{Th})^{3/2}}{\sqrt{2\ \rho\ \mathrm{A_R}}} = \frac{(\mathrm{Th})^{3/2}}{\sqrt{\frac{\pi}{2}\rho\ \mathrm{D_i}}}. \tag{15}$$

Since each rotor should have the same diameter, using (14), the total sum of power required ($\mathrm{P_{TOT}}$) for a specific multirotor model can be obtained in the following fashion:

$$\mathrm{P_{TOT}} = \mathrm{N_R}\ \frac{(\mathrm{Th})^{3/2}}{\sqrt{\frac{\pi}{2}\rho\ \mathrm{D_i}}} = \frac{(\mathrm{N_R}\ \mathrm{Th})^{3/2}}{\sqrt{\frac{\pi}{2}\rho\ \mathrm{N_R}\ \mathrm{D_i}}} = \frac{\mathrm{W}^{3/2}}{\sqrt{\frac{\pi}{2}\rho\ \mathrm{N_R}\ \mathrm{D_i}}}. \tag{16}$$

In the hypothetical case multirotors with different number of rotors had similar weight and propeller diameter, the model with more rotors would require less power to be kept in hover mode and have an increased flight time. However, in a realistic situation, increasing this value inherently raises the frame's weight and is not as effective as using larger propellers. An interesting property featured in equation (16) is how total power grows asymptotically to $\rho^{-1/2}$, like the equation derived for fixed-wing aircraft.

Equations (13) and (16) represent the power that would be required by fixed-wind and multirotor UAVs respectively to fly at a certain altitude. Both are affected by weight in a similar manner and, when comparing power at different altitudes (point 1 being lower than point 2) and assuming the rest of the variables remain unchanged, the following proportion can be established for both cases:

$$\frac{P_{R2}}{P_{R1}} = \sqrt{\frac{\rho_1}{\rho_2}}. \tag{17}$$

Using equation (6), the increase in required power for both UAV configurations relative to sea level conditions is displayed in Figure 5. Comparing both graphs in Figure 2 and Figure 5, required power increases less than air density decreases up to the established limit height.

These equations can be used to estimate the effect of design parameters on power consumption of both UAV configurations and to correlate propeller and aircraft power

requirements. Before proceeding, a clarification should be made the rotors' work regime in fixed-wing and multirotor frames. Under a certain set of circumstances, such as low wind speed or when flying along a wind current, there will be instances in which the propeller of a fixed-wing small UAV won't have to exert thrust to maintain altitude and will require almost no energy, so battery consumption may be irregular depending on the chosen flight path. In contrast, to maintain hover conditions, the propellers of a multicopter will have to keep rotating at a certain frequency to generate enough thrust to counter the weight of the frame, meaning electrical energy consumption will remain almost constant.

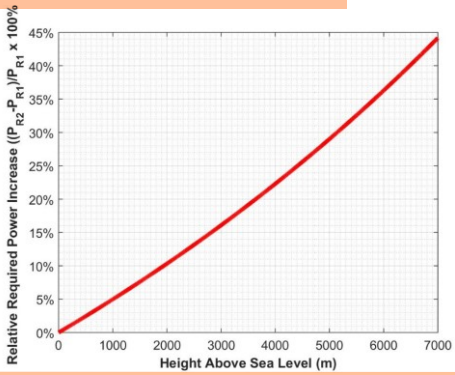

Fig. 5.- Increase of power required by UAVs relative to sea level with altitude increase.

III. Flight Tests' Frames and Sites

The frames used in this study were outfitted in PUCP's laboratories for high altitude applications. A flying wing aircraft was chosen to represent fixed-wing frames and a quadcopter model does the same for multirotors. The data displayed in section IV was obtained from flight tests carried out to assess the frame's performance and data gathering exercises.

The fixed-wing UAV was based on the Skywalker-X8 frame and, depending on the assigned task, it can be outfitted with several sensors, although most times it is used to carry cameras. The size of the frame (2.12 meters wingtip to wingtip and an approximate wing area (S) of 0.7 m^2) allows it to generate enough lift to carry relatively heavy payloads (up to 2 kg) and keep a stabilized flight in the presence of strong winds. A folding propeller was attached to this model, so that it folds during landing operations to prevent any damage. The UAV is showcased in Figure 6.

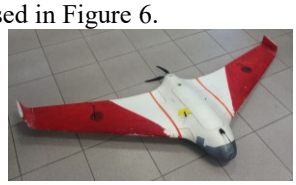

Fig. 6.- X8 - Skywalker UAV model for multipurpose appliances.

The quadcopter featured in the tests was developed by qAIRa, a drone startup that operates at PUCP, to monitor operations and activities at high-altitude mining venues. Quadcopter frames have simple dynamics (which make them easy to control) and the space between arms (820 mm rotor to rotor) allows for the usage of large diameter propellers. qAIRa's model was created by modifying a Tarot X4 frame and features two different sets of carbon fiber propellers to use in either low or high-altitude environments. Furthermore, it employs a couple of high energy batteries (Max Amps® 11 Ampere-hour 6S 40 C) and energy efficient motors (Tiger Motor® U7) to enhance the UAV's flight time. The multirotor is shown in Figure 7.

Fig. 7.- qAIRa's Quadcopter for Monitoring of High-Altitude Mining Facility

Flight tests were held at high-altitude environments as well as near sea level terrain. At high-altitude locations, while fixed-wing aircraft operations required at least an extension of land to comfortably perform landing operations, multirotors could take-off and land from any flat surface. The quadcopter model was flown at the Campomar U model aircraft flight site (65 MASL), Antamina (4240 MASL) and Ticlio (4840 MASL). The flying-wing model operated above Ica Farms (585 MASL), the Huaraz Highlands (3650 MASL), the Suyuparina Glacier (5610 MASL) and the Ubinas Volcano (5830 MASL).

IV. Flight Results and Data Discussion

Figure 8 shows flight data from two missions in which the flying wing UAV was used, one held at the Huaraz Highlands and the other over the Suyuparina Glacier, both with the purpose of taking aerial images. These graphs compare the missions' altitude, the current required by the UAV at each time (which represents the required power) and the accumulated current, which represents summation of all previous current values up to a point in time. The Suyuparina Glacier required more power for a flight of similar duration as demonstrated by the accumulated current. It should be noted that peaks in measured current are displayed while performing take-off, climbing and landing maneuvers (if landing speed is too large, the procedure is aborted and altitude must be gained before trying again). Another interesting fact to point out is that during the Huaraz flight no energy was consumed at times as shown in the corresponding current graph, which was a possibility that was mentioned in the last paragraph of section II and could be explained through wind flow or other factors.

Figure 9 in the following page shows the same information as Figure 8 for the quadcopter model displayed in section III. Similarly to the fixed-wing scenario, higher altitudes demand higher power. The Antamina Mine site flight required almost as twice total current than the Campomar U one, which can't be solely attributed to the near 6-minute flight time difference or the different propellers used in both flights. As established before, to maintain a certain altitude, the quadcopter required an almost continuous amount of current; this is further reinforced by having linear accumulated current graphs. Like the other case, current peaks are experienced by multirotors during climb operations, although landing does not necessarily require high current. The difference between the

last values of the accumulated current graphs of the fixed-wing and quadcopter modes is notorious because, in practice, multirotors will require higher energy batteries than fixed-wing UAVs to fly at the same altitudes.

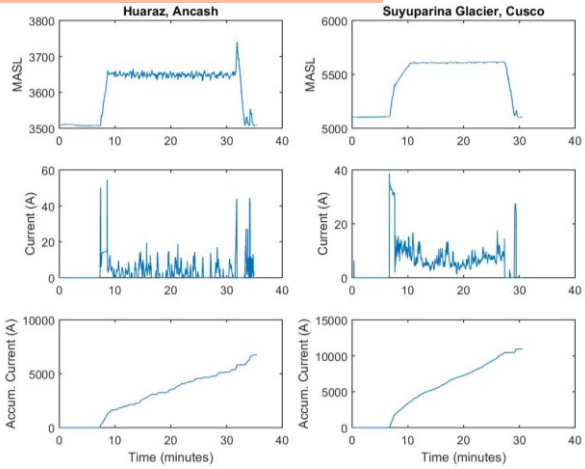

Fig. 8.- Fixed-Wing UAV flight data comparison.

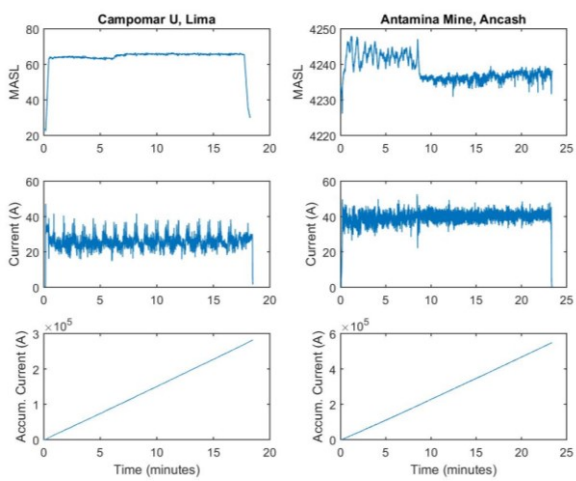

Fig. 9.- Quadcopter UAV flight data comparison

The validity of equation (17) is evaluated using Figure 10 as a reference. Air density values were obtained from equation (6) by using the mean mission altitude as input. Power values were obtained by calculating the average power used during these missions at the cruise phase at the target altitude. Most of the flight tests evaluated accommodate near the predicted ratio, although the fixed-wing UAV presents some noticeable outliers. The red plus sign can be explained by the circumstances under which the Suyuparina flight was held. Not only was this flight's target altitude higher, the winds around the area were turbulent and required a higher thrust to allow regular flight speed, which is why the marker is higher than the red linear prediction. The cyan and black plus signs can be elucidated by considering the temperature at the Ica test site, which is higher than the standard one assumed, resulting in lower air density than expected and higher power requirements. Equation (6) doesn't account for temperatures that differ from standardized assumptions and creating an expression that included said variables would require a more complex analysis.

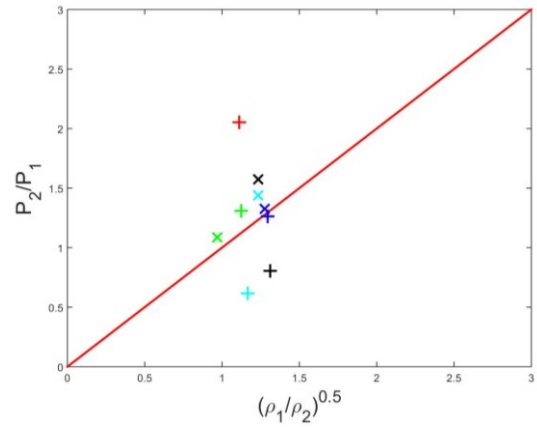

Fig. 10.- Graph for equation (17) validation. The red line represents the y = x linear function. Table II shows the comparisons represented by each point.

TABLE II
COMPARISON BETWEEN FLIGHT TESTS IN FIGURE 10

UAV Type	Site 1	Site 2	Marker	Color
Fixed-Wing	Huaraz	Suyuparina	+	Red
	Huaraz	Ubinas		Green
	Ica	Huaraz		Cyan
	Ica	Suyuparina		Blue
	Ica	Ubinas		Black
Multicopter	Ticlio	Antamina 1	×	Green
	Campomar	Ticlio		Blue
	Campomar	Antamina 1		Cyan
	Campomar	Antamina 2		Black

This analysis shows that non-ideal flying conditions might disturb predicted power demands and reduce estimated flight times. Further tests will have to be performed to start using equation (17) as an accurate prediction tool, but, for the time being, it should prove useful to estimate UAV performance. Through the information presented in this document and the flight test experiences, a few guidelines have been established for both high-altitude multicopter and fixed-wing UAV design. For multicopters, the following guidelines should be considered:

- Larger diameter propellers will generate more lift and perform better at higher altitudes if the motor and battery configuration allows it, since it will require more power as well due to inertia. Efficient motors for UAV applications exist but are usually expensive.
- If the multicopter is using large propellers, high voltage batteries will be required for acceptable multirotor stabilization, since high voltages allow these rotors to change rotating speed fast enough to react to sudden vibrations.
- Having more rotors will yield better stabilization and may even help reduce battery consumption. This also diminishes the space available for propellers, so careful consideration must be taken when evaluating the geometrical constraints of the design. Furthermore, having more rotors usually results in more weight, which may increase energy consumption and decrease flight time.
- High-endurance multirotors usually feature high-voltage, high-energy batteries, and high-altitude ones will need these as well. However, these are often voluminous and heavy, and may even be counterproductive without the correct set of propellers and motors.

As for fixed-wing UAVs, the suggested guidelines are the following:

- Most fixed-wing UAVs will have acceptable flight times at high-altitude locations given a successful take-off; however, depending on the payload and travel distance, different configurations will work better than others.
- Frames with large wings perform better at higher altitudes since they possess greater surface area which eases lift generation, although some special arrangements must be made for their transportation due to their size.
- Flying-wing frames perform well in high altitudes since most of their surfaces generate lift and relatively large-sized models can carry somewhat heavy payloads. Nonetheless, maneuverability is more limited due to the lack of a tail and the flight controller may require more careful tuning for acceptable performance.
- Medium-sized frames with tails have great maneuverability, take turns better than their tail-less counterparts and are well-suited to carry low-weight payloads at high altitudes.
- As with multicopters, batteries with high amounts of Ampere-hour (in the case of LiPo batteries) will extend the flight time of fixed-wing models. Consequently, the frames will have to be capable of generating enough lift to compensate for the extra weight. Furthermore, the battery position must be carefully chosen since it can affect overall maneuverability by changing the frame center of mass.
- Whether a frame features landing gear will depend on the flight mission location. Take-off and landing operations are dangerous for aircraft with landing gear if a runway with proper clearance and extension isn't available, which is frequent at places where high-altitude small-sized UAV operations are performed. Moreover, these increase weight and drag (unless they can be retracted).
- Retractable propellers may be preferred if the frame rotor is located at body level, where landing can cause some damage if landing gear isn't available.

V. Conclusions and Future Work

The effect of high-altitude environments and low air-density on multicopter and fixed-wing UAVs was studied in this paper through theoretical background review and substantiated by analyzing flight test logs. Energy requirements increase as target altitude does, which decreases flight time; these can also be influenced by unforeseeable factors, like wind turbulence and temperature. Proper aeronautic design and choosing the right combination of propulsion elements will aid a UAV to fare better under these circumstances, which is crucial for industrial applications.

Fixed-wing UAVs are suited for 3D reconstruction and mapping applications. Plane configurations offer maneuverability and handle small and low-weight payloads well. Large-sized flying-wing configurations are less maneuverable but generate more lift, can handle relatively heavy payloads and perform better at higher altitudes. Multicopter UAVs perform better when positioning in 3D environments is needed, such as in surveillance tasks and sensor data recollection. Ideally, higher rotor quantity will increase stability and larger propeller diameter will ease weight lifting tasks at higher altitudes. However, larger propellers require high voltage batteries to allow motors to change rotation speed fast enough to keep the UAV stable.

Future work will focus on continuing developing unmanned aircraft for high-altitude applications and the resulting guidelines will focus more on specific applications. Also, more data, including temperature, will be obtained to find a more fitting correlation between energy consumption, altitude and temperature, along other possible variables that may affect flight performance considerably to attempt to predict UAV behavior in different altitude from initial flight tests at a given altitude.

Acknowledgements

Our special thanks to the startup qAIRa, the SANT (Unmanned Aerial Systems) and the GIT (Technological Innovation Group) investigation groups from the Pontifical Catholic University of Peru for the data provided for this paper's analysis.

References

[1] Federal Aviation Administration, "Fact Sheet – Small Unmanned Aircraft Regulations (Part 107)", 21-Jun-2016. [Online]. Available: https://www.faa.gov/news/fact_sheets/news_story.cfm?newsId=20516. [Accessed: 8-Febr-2017]

[2] EASA, "Civil drones (Unmanned aircraft)". [Online]. Available: https://www.easa.europa.eu/easa-and-you/civil-drones-rpas. [Accessed: 25-May-2017].

[3] DJI, "Matrice 600 Pro Specs". [Online]. Available: http://www.dji.com/matrice600-pro/info#specs, [Accessed: 10-Feb-2017]

[4] Microdrones, "The heavy-lifting drone: MD4-3000". [Online]. Available: https://www.microdrones.com/en/mdaircraft/md4-3000/. [Accessed: 10-Febr-2017]

[5] senseFly, "eBee: The professional mapping drone". [Online]. Available: https://www.sensefly.com/drones/ebee.html. [Accessed: 10-Feb-2017].

[6] senseFly, "Mapping the Matterhorn: senseFly SA". [Online]. Available: https://www.sensefly.com/user-cases/mapping-the-matterhorn.html. [Accessed: 12-Feb-2017].

[7] I. Guzmán, "La PUCP investiga con drones a más de 5 mil metros de altura," *Punto Edu*. [Online] Available: http://puntoedu.pucp.edu.pe/videos/exploracion-en-las-alturas/, [Accessed: 10-Feb-2017].

[8] D. Longo, D. Melita, G. Muscato and S. Sessa, "A Mixed Terrestrial Aerial Robotics Platform for Volcanic and Industr.ial Surveillance" in *Safety, Security and Rescue Robotics*, Rome, 2007.

[9] K. Nagatani, Y. Hada, S. Yuta, K. Fujino, T. Izu, R. Mackay, "Volcanic Ash Observation in Active Volcano Areas using Teleoperated Mobile Robots", in *Safety, Security, and Rescue Robotics (SSRR)*, 2013.

[10] RPP Noticias, "Moquegua: dron sobrevuela cráter del volcán Ubinas," 19-Aug-2016. [Online]. Available: http://rpp.pe/peru/moquegua/moquegua-dron-sobrevuela-crater-del-volcan-ubinas-noticia-988549. [Accessed: 10-Feb-2017].

[11] Prof. Z. S. Spakovszky, "Performance of Propellers," *Thermodynamics and Propulsion*. [Online]. Available: http://web.mit.edu/16.unified/www/SPRING/thermodynamics/notes/node86.html. [Accessed: 10-Feb-2017].

[12] Gerrit J.J. Ruijgrok, "The Atmosphere" in *Elements of Airplane Performance*, 2nd edition, Delft, The Netherlands: VSSD, 2009.

[13] Gerrit J.J. Ruijgrok, "The airplane in symmetric flight" in *Elements of Airplane Performance*, 2nd edition, Delft, The Netherlands: VSSD, 2009.

[14] Gerrit J.J. Ruijgrok, "Performance in steady symmetric flight" in *Elements of Airplane Performance*, 2nd edition, Delft, The Netherlands: VSSD, 2009.

[15] M. Bangura, M. Melega, R. Naldi, and R. Mahony, "Aerodynamics of Rotor Blades for Quadrotors," 05-Jan-2016. [Online]. Available: https://arxiv.org/abs/1601.00733#. [Accessed: 10-Feb-2017].